UNIVERSITY COLLEGE OF ENGINEERING (A)
Osmania University, Hyderabad-07

CLASS TEST-II

Subject: Operating Systems

Class: B.E (CSE)
Semester: IV
Academic Year: 2024-2025

Max Marks: 30
Duration: 12:00 noon to 01:30 p.m.)
Date: 26-06-2025

S. No.	PART A Answer All Questions (5 x 2 = 10 Marks)	Marks	BT	CO
1.	Differentiate paging and segmentation.	2	2	1
2.	List out the major attributes and operations of a file system.	2	1	1
3.	What are the advantages and disadvantages of using an access matrix.	2	2	2
4	If the size of logical address space is 2 to the power of m, and a page size is 2 to the power of n addressing units, then the high order _____ bits of a logical address designate the page number, and the ____ low order bits designate the page offset. (A) m, n (B) n, m (C) m – n, m (D) m – n, n	2	1	1
5	The LRU algorithm (A) pages out pages that have been used recently (B) pages out pages that have been least used recently (C) pages out pages that have not been used recently (D) pages out the first page in a given area	2	2	2
	PART B **Answer Any Two Questions (2 x 10 = 20 Marks)**			
6.	Define demand paging in memory management. What are the steps required to handle a page fault in demand paging?	4	1	3
	Consider the following page reference string: 7,2,3,1,2,5,3,4,6,7,7,1,0,5,4,6,2,3,0,1 Assuming the demand paging with three frames, how many page faults would occur for the following replacements algorithms? a. LRU b. FIFO c. Optimal	6	2	2
7.	Write a short note on the actions to be performed during a file deletion operation if link exist in the FS directory hierarchy.	5	2	2
	Describe with a neat sketch about the various directory structure.	5	1	1
8.	How is an Access Matrix implemented in an operating system? Explain.	5	2	2
	Explain Banker's Algorithm with an example.	5	1	3